

SQL Joins ✓

id	name
1	Abrar
2	Bashir
3	Cassie
4	Dylan
5	Erin
6	NULL

Students

⋈

id	hobby
2	Fishing
1	Singing
3	Gardening
1	Gaming
5	Sleeping
9	NULL

Hobbies

1. Inner Joint



2. Left Joint



3. Right Joint



4. Full Joint



Format: `SELECT s.id, s.name, h.hobby FROM student s JOIN Hobbies h ON s.id = h.id;`

1. Inner Join:

```
MariaDB [jointest]> select t1.name, t2.hobby from students t1 join hobbies t2 on t1.id = t2.id;
```

name	hobby
Bashir	Fishing
Abrar	Singing
Cassie	Gardening
Abrar	Gaming
Erin	Sleeping

5 rows in set (0.026 sec)

2. Left Join:

```
MariaDB [jointest]> select t1.name, t2.hobby from students t1 left join hobbies t2 on t1.id = t2.id;
```

name	hobby
Bashir	Fishing
Abrar	Singing
Cassie	Gardening
Abrar	Gaming
Erin	Sleeping
Dylan	NULL
NULL	NULL

7 rows in set (0.000 sec)



3. Right Join:

```
MariaDB [jointest]> select t1.name, t2.hobby from students t1 right join hobbies t2 on t1.id = t2.id;
```

name	hobby
Abrar	Singing
Abrar	Gaming
Bashir	Fishing
Cassie	Gardening
Erin	Sleeping
NULL	NULL

5 rows in set (0.000 sec)

4. Full Join:

```
MariaDB [jointest]> select t1.name, t2.hobby from students t1 left join hobbies t2 on t1.id = t2.id  
-> union  
-> select t1.name, t2.hobby from students t1 right join hobbies t2 on t1.id = t2.id;
```

name	hobby
Bashir	Fishing
Abrar	Singing
Cassie	Gardening
Abrar	Gaming
Erin	Sleeping
Dylan	NULL
NULL	NULL

7 rows in set (0.001 sec)

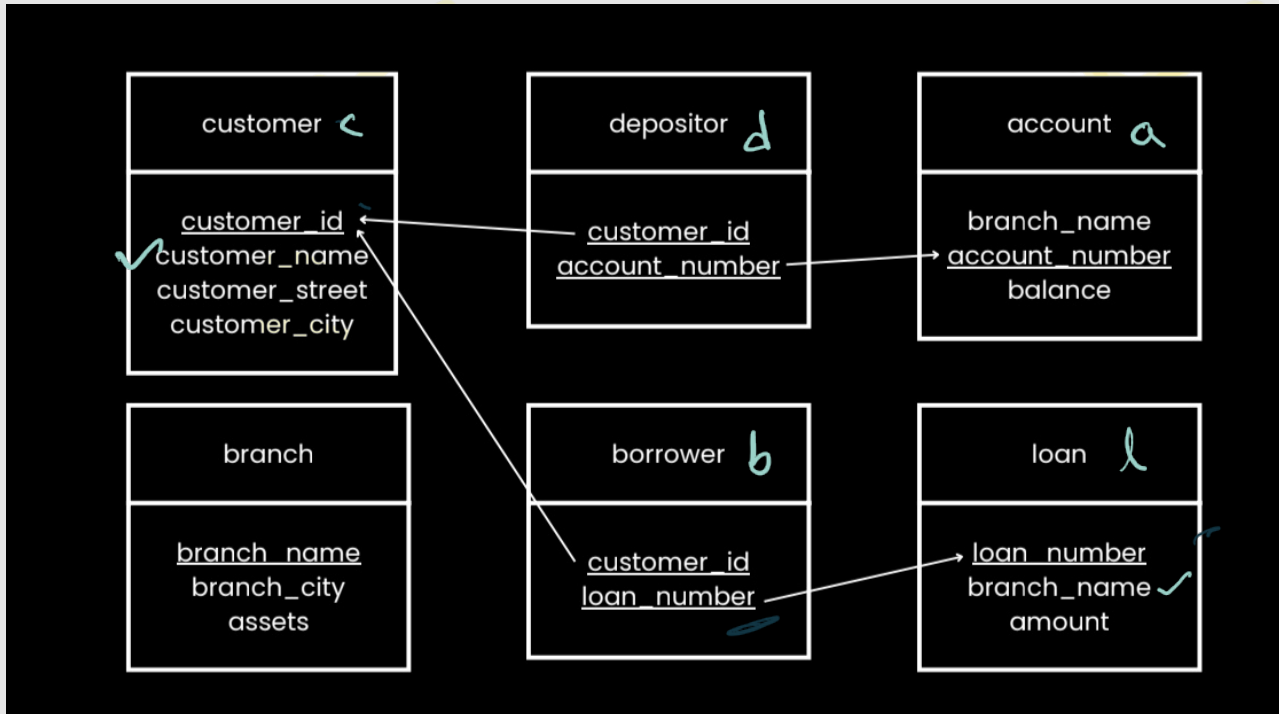


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Question: (a) Join customer name, city, account number, balance and branch name

(b) Solve without join keyword

(c) Find name of customer who have loan in Mirpur branch



(a) SELECT c.customer_name, c.customer_city,
d.acc-no, a.branch_name, a.balance

→ FROM customer c

→ JOIN depositor d ON c.customer_id = d.customer_id

→ JOIN account a ON d.acc-num = a.acc-num;

(b) SELECT c.customer_name, c.customer_city,
d.acc-no, a.branch_name, a.balance

→ FROM customer c, depositor d, account a

→ WHERE c.cus_id = d.cus_id

→ AND d.acc-num = a.acc-num;

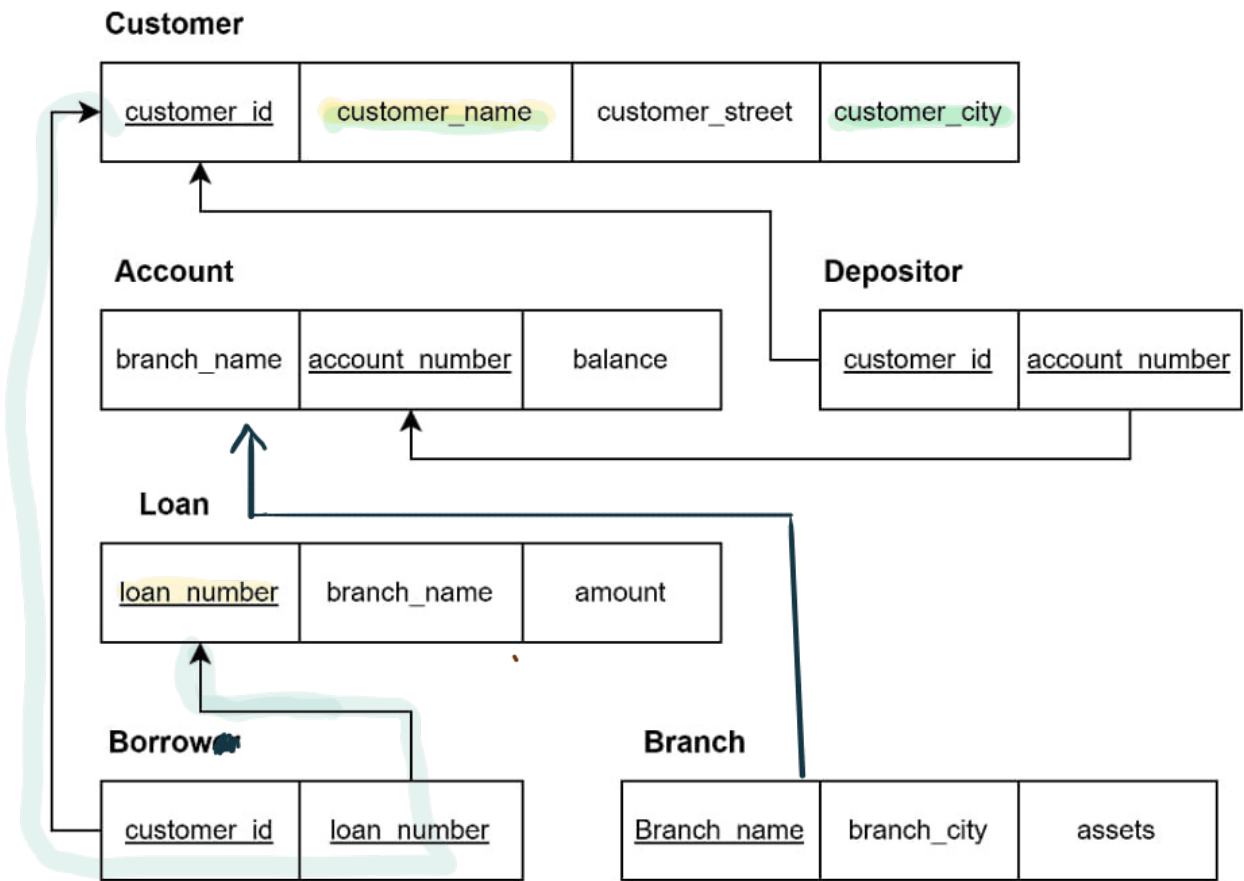
(c) SELECT c.name FROM customer c

→ JOIN borrow b ON c.cus_id = b.cus_id

→ JOIN loan l ON b.loan-no = l.loan-no

→ WHERE l.branch_name = 'Mirpur';

BANK DB



(a) Find name of customer who has taken loan

- * (b) Find name of customer who has **not** taken loan
- * (c) Find **pair** of customers who are from the same city.
Format: Customer1, Customer2, City.

(a) `SELECT c.cus_name FROM customer c`

→ `JOIN borrow b ON c.cus_id = b.cus-id`

→ `JOIN loan l ON b.loan-no = l.loan-no;`

(b) `SELECT name FROM customer WHERE
id NOT IN (SELECT customer-id FROM
borrow);`

(c) `SELECT a.cus_name AS cus1,`

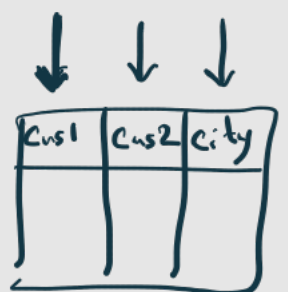
→ `b.cus_name AS cus2,`

→ `a.city AS city`

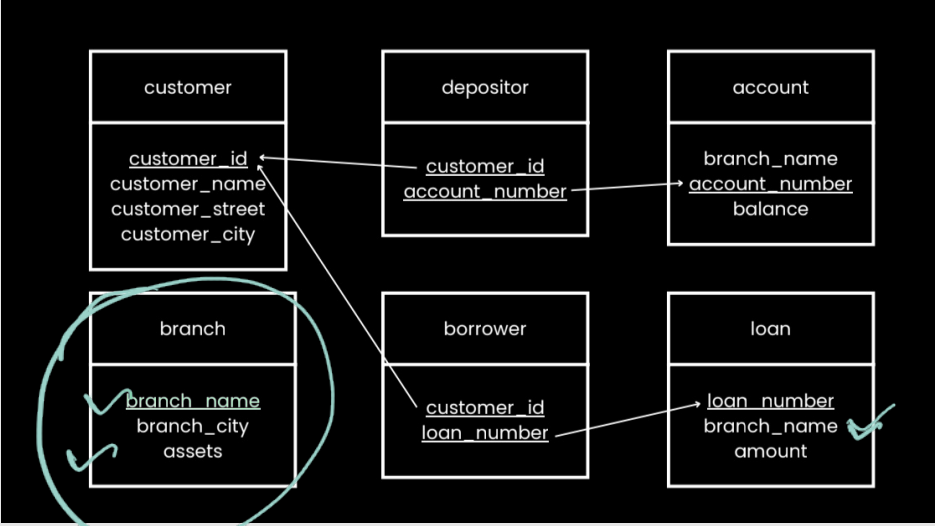
→ `FROM customer a`

→ `JOIN customer b ON a.cus-city = b.cus-city`

→ `AND a.customer-id > b.customer-id;`



Question : Find name of branch whose assets are greater than the assets of some branch in Uttara



SELECT name FROM branch WHERE asset >
→ ANY
→ (SELECT asset FROM branch WHERE city = 'Uttara');

SQL JOINS

like '%...'

Task 6.3:
Find the names of customers on streets with names ending in "Hill"
-> select customer_name, customer_street from customer where customer_street like '%Hill';

customer_name
Glenn

SQL JOINS

Task 6.6:
Find the set of names of customers at Brighton branch, in alphabetical order
-> select c.customer_name from customer c, depositor d, account a where c.customer_id = d.customer_id and a.account_number = d.account_number and a.branch_name = 'Brighton' order by c.customer_name;

customer_name
Johnson
Jones

SQL JOINS

Task 6.7:

Show the loan data, ordered by decreasing amounts, then increasing loan numbers

-> select * from loan order by amount desc, loan_number asc;

loan_number	branch_name	amount
L-23	Redwood	2000
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-11	Round Hill	900
L-93	Mianus	500

SQL JOINS

*

Task 6.9:

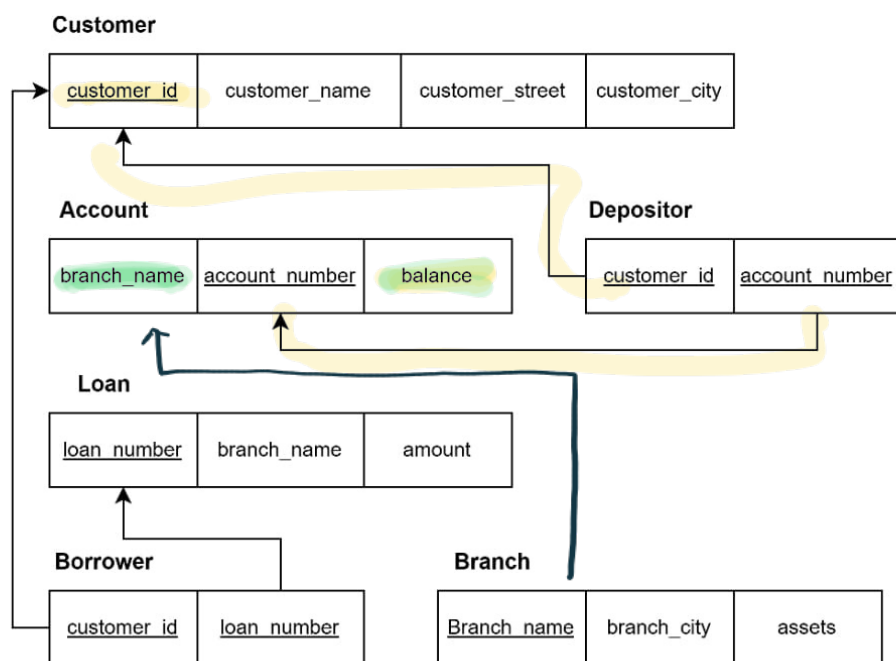
Find the names and account number of customers who have the 3 highest balances in their accounts

-> select c.customer_name, a.account_number, a.balance from customer c, depositor d, account a where c.customer_id = d.customer_id and a.account_number = d.account_number order by a.balance desc limit 3;

Highest
↳
Limit 3

customer_name	account_number	balance
Johnson	A-201	900
Jones	A-217	750
Smith	A-215	700

BANK DB



2	100
3	300
2	100

id	total

* (a) Find customer_id and sum of balances of all customer who have more than one account. (Display sum as total)

SELECT c.customer_id, Sum(a.balance) AS total

→ FROM customer c

→ JOIN depositor d ON c.cus-id = d.dep-id

→ JOIN account a ON d.acc-no = a.acc-no

→ GROUP BY c.id HAVING COUNT(d.acc-no) > 1;

$\text{ROUND}(\text{AVG}(\text{balance}), 3)$

(b) Show 4% of balance as interest (round up to 2 digit)
* across each branch

```
SELECT branch_name, ROUND(SUM(balance * 0.04), 2)
AS interest FROM account
→ GROUP BY branch_name;
```



```
create table customer (
customer_id varchar(10) not null, ← E.I.C
customer_name varchar(20) not null,
customer_street varchar(30),
customer_city varchar(30),
primary key (customer_id)); ← K.C
```

```
create table branch (
branch_name varchar(15),
branch_city varchar(30),
assets int,
primary key (branch_name),
check (assets >= 0)); ← Domain
```

✓ create table depositor (
customer_id varchar(10) not null,
account_number varchar(10) not null,
primary key (customer_id, account_number),
foreign key (customer_id) references customer(customer_id),
foreign key (account_number) references account(account_number));

customer

